NARRATIVE QUARTERLY REPORT
(APRIL–JUNE) 2025

Project — In situ monitoring of water quality in Cuban bays: creating and promoting the use of scientific tools

Participating institutions

- University of Havana (UH)
- Université Libre de Bruxelles (Belgium)

Cooperating institutions

- Center for Environmental Research and Transport Management (CIMAB)
- ESPOLETA Tecnologías (SME)

Other collaborating institutions

- José Antonio Echeverría Technological University of Havana (Cujae)
- Havana Bay State Management Group
- Université Catholique de Louvain

Project coordinators

Cuban side: Dr. C. Ernesto Altshuler Álvarez
Foreign side: Dr. Denis Terwagne

Participants — Dr. Geydis Fundora Nevot, Dr. Florence Degavre, Dr. Stamatios Nicolis, Dr. Alexandre Campo, Lic. Daniela Martínez Ortiz, Eng. Alex Rivera Rivera, MSc. Luis Abel Rodríguez de Torner, Dr. Aramis Rivera Denis, Eng. Orestes Chávez Linares.

Duration — Five years

General objective — To provide the Cuban scientific-social ecosystem with a set of tools that enables in situ monitoring of water-quality parameters in Cuban bays, in a sustainable way and connected to local communities.

Specific objectives

- Design and build a wave channel at the Faculty of Physics, UH
- Design and build a buoy prototype
- Set up a server with a website to receive data from the buoys and from the community
- Establish a methodology for using the buoys
- Train the corresponding human resources

Fulfillment of the quarterly objectives

Objective 1. Drafting the second-year purchase list and customs procedures

Work continued on the complex permits needed to clear from Customs a box containing an alarm system for our laboratories, which remains held there. As one step in that process, a specialist from the authorized institution carried out the alarm-placement design, visiting our facilities and discussing with us the best options for positioning the security devices. During the last quarter, constant negotiation continued with suppliers of specialized equipment, such as a UV-visible spectrophotometer, a high-speed camera, water-recirculation systems for the wave channel, and others.

Objective 2. Continued instrumentation of the buoys, and wave-channel design

Throughout the period, multiple tests were run on the buoys' LoRa-based communication system, involving PhD candidates Alex Rivera and Daniela Martínez. In particular, the resilience of the communication system with 12-volt batteries was tested on land to minimize the effect of the country's systematic blackouts. During his second stay at the Université Libre de Bruxelles starting in February 2025, PhD candidate Luis Abel Rodríguez tested more effective ways to build medium-sized channels, investigating a variety of adhesives and assembly systems. He also ran relevant experiments, culminating in a results-discussion seminar held on June 27, 2025, attended, besides Luis Abel, by the Belgian counterparts and E. Altshuler via videoconference.

Detail of the generator inside the wave channel built by PhD candidate Luis Abel Rodríguez during his stay at ULB in the first months of 2025.

A parallel effort to Luis Abel's, on image processing in the channel (specifically for tracking empty beverage cans that pollute Cuban marine waters), was carried out by Physics student Jean Rafael López, earning him second place in the Electronics and Computing category of the Faculty of Physics' Student Scientific Conference at the University of Havana.

Objective 3. First steps in developing a wireless sensor for seawater conductivity

Physics graduate Marlon Mederos continued tests, both in the lab and in Havana Bay waters, on the possibility of using LoRa signal-reception strength as a wireless measure of water conductivity/salinity. The goal is to avoid the standard method of using metal electrodes in direct contact with the water, which would require a high level of maintenance. Preliminary tests have not yet produced conclusive results.

Objective 4. Academic exercises carried out by the PhD candidates involved in the Project

Formal steps were taken to enroll PhD candidates Luis Abel Rodríguez and Alex Rivera in the doctoral program at the Université Libre de Bruxelles. In addition, the first steps were taken in writing a review article on the use of artificial intelligence for detecting biological marine contaminants, whose lead author is Lic. Daniela Martínez.

Objective 6. Continued progress in social-science research

One strand of the research is assessing the expectations and needs of social actors regarding the development of scientific tools for continuous water-quality monitoring, as well as access to relevant environmental-management data. Toward that end, this quarter saw the first meeting with fishers based at the Casablanca recreational fishing base. The project's general outline was presented, along with the importance of their participation and the benefits of the new technology for their activity. The fishers said this is the first time a project has directly involved them.

First meeting with fishers based at the Casablanca recreational fishing base, held on April 9, 2025. At center, Lic. Mayelín García Remus.
Exchange with the robotics project at the "La Colina" Sociocultural Center. Demonstration of drinking-water quality measurement.

Other experiences shared by the fishers included: (a) they use the Windy app to track weather variables such as wave height and wind direction; (b) they consider it important to have new technology that lets them know parameters about the bay's water, mentioning tides, wind direction, salinity, water temperature, and pollution level — on this last point, they raised the need to know the degree of oil contamination; (c) the bay's main pollution comes from oil and from garbage carried by the rivers, especially during heavy rain, and due to this pollution many fish no longer come into the bay; (d) they do not fish inside the bay itself, since it is prohibited — they fish offshore, roughly 4 to 7 miles out; those who fish inside the bay are unlicensed, illegal fishers; (e) they suggested placing the buoys somewhere they themselves could keep an eye on them. Actor mapping continued for the key-informant technique, along with designing the Green Map and the community workshops to be held next quarter; the project was presented to key actors with emphasis on the activities and objectives involving the Casablanca community. Informants contacted and briefed this quarter include the Intendente of Regla municipality, the Director of the Local Development Office, the Port Captaincy's Casablanca sub-station, and the "La Colina" Sociocultural Center, which hosts a robotics project with secondary-school students that includes digital fabrication and sensor design for drinking-water quality measurement, complementing this project's focus on Havana Bay's waters.

An article on this exchange with these actors is being prepared for the "Diálogos" bulletin of the Latin American Council of Social Sciences (CLACSO). Another research strand in the social sciences is systematizing monitoring, training, communication, and environmental-management practices related to water quality, in order to build a typology of practices as a baseline for identifying the social innovations spurred by participation in this project. On this front, progress was made compiling and analyzing research practices on environmental monitoring and sanitation in Havana Bay carried out by CIMAB from the 1980s to the present — including, among others, the Regional Project CUB 80/001 (UNDP-UNEP-UNESCO) "Research and control of pollution in Havana Bay" (1980–1984); the Havana Bay monitoring system, "Periodic control of the bay ecosystem's quality" (1986–1995); the GEF/UNDP project "Environmental planning and management of heavily polluted bays and coastal zones of the Greater Caribbean: Regional Project, Havana Bay, Cuba case study" (1996–1998); the Havana Bay Surveillance System (1996–present); and a long list of environmental-license requests and diagnostics carried out between 2000 and 2010. With the results of this systematization, the paper "Environmental monitoring and management in community transformation: experiences from Casablanca" is being drafted, to be presented at the 16th International Workshop "Communities: History and Development 2025," within the 5th UCLV International Scientific Convention 2025 at the Central University "Marta Abreu" of Las Villas.

Another advance is the methodological design of the thematic Green Map workshops (environmental management, actors, and innovation processes in the Casablanca community) to be held in September 2025; toward that end, two project members were nominated for a training experience, a Casablanca case-study file was prepared, and the project took part in the introductory session of UCLouvain's "Circle U. PReCISE Summer School" on citizen science.

Introductory session of the virtual "Circle U. PReCISE Summer School," June 23, 2025.

Sincerely,
Dr. C. Ernesto Altshuler
Head of the Complex Systems Laboratory, Center for Complex Systems, Faculty of Physics
University of Havana.

Machine-assisted English translation of the original Spanish narrative report submitted to ARES. Figures and photographs are reproduced in the original Spanish document.